

RESEARCH ARTICLE

Comparison of Evolutionary and Traditional Algorithms for Flight Route Optimization

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ABSTRACT This study investigates flight route optimization in a large airline network by comparing traditional path-finding algorithms with an evolutionary approach under realistic constraints. Using a real-world dataset of 6,817 routes, Dijkstra's algorithm, the A* search algorithm with precomputed heuristics, and a Genetic Algorithm (GA) were implemented. All methods were evaluated on identical origin–destination pairs while enforcing an exact number of layovers, and on scenarios where direct flights were removed or newly introduced. The results show that, although Dijkstra's algorithm always returns the optimal shortest-time route when one exists, its computation time increases rapidly as the number of required layovers increases. A* improved the efficiency in dense subgraphs but sometimes failed under strict layover constraints owing to heuristic misalignment. In contrast, GA consistently produced near-optimal routes with stable runtimes across varying connectivity levels and constraint settings. Route removal experiments highlighted the vulnerability of sparsely connected regions, whereas adding direct flights mainly benefited airport pairs previously linked only by long multi-stop paths. These findings indicate that evolutionary algorithms are well-suited for large-scale, dynamically changing airline networks, whereas classical algorithms remain preferable for smaller, well-structured subproblems.

INDEX TERMS Flight route optimization, genetic algorithm, Dijkstra's algorithm, A* search algorithm, transportation networks, evolutionary algorithms, route planning, layover constraints, dynamic optimization.

I. INTRODUCTION

The vehicle routing problem (VRP) is one of the most long-standing and computationally demanding problems in transportation and logistics. Originally introduced by Dantzig and Ramser in 1959 [1], the VRP serves as a fundamental framework for determining efficient vehicle routes that minimize operational costs, travel time, and fuel consumption. Its relevance has continued to grow as the scale and interconnectedness of modern transportation systems have increased, requiring routing problems to be solved under realistic, often dynamic constraints. Traditional deterministic algorithms provide effective solutions for small and moderately sized instances; however, their performance typically deteriorates as network complexity increases. In contrast, evolutionary algorithms, particularly genetic algorithms (GAs), inspired by the principles of natural selection, offer a more adaptive

and scalable alternative capable of evolving high-quality solutions for large and complex optimization tasks.

Flight route optimization exemplifies the challenges inherent in VRP. Airlines must determine efficient paths between airports while accounting for operational costs, travel times, fuel usage, regulatory constraints, delays, and schedule variability. As the number of airports and available connections increase, identifying an optimal route becomes increasingly difficult. The problem is further compounded by real-world factors, such as air traffic regulations, route adjustments, and fixed layover requirements, all of which influence network performance and operational efficiency. Consequently, ineffective routing decisions can lead to extended travel times, higher costs, and suboptimal utilization of network resources.

To address these challenges, this study investigated flight route optimization by examining both traditional path-finding algorithms and evolutionary approaches. Specifically, it analyzes Dijkstra's algorithm and A* Search as classical deterministic methods and compares them with a genetic

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algorithm capable of exploring complex solution spaces. This study incorporates constraints such as a fixed number of layovers and evaluates how each algorithm responds to changes in the network structure, including the removal or introduction of direct routes. The following questions guide this research:

- RQ1. How does the removal of specific routes influence the network efficiency when the number of layovers between the origin and destination is fixed?
- RQ2. How does the introduction of a new direct flight affect performance in terms of travel time and potential profitability relative to the existing indirect routes?
- RQ3. Which class of algorithms, traditional or evolutionary, produces optimal routing solutions under dynamic network conditions, including constraints on the number of stopovers?

Although Dijkstra's algorithm, A* Search, and GA have been extensively studied, their comparative performances under explicit layover constraints and dynamic network modifications remain insufficiently explored. This study addresses this gap by providing methodological insights and practical guidance for airline route planning. The findings extend beyond aviation and contribute to broader transportation network optimization, offering stakeholders a clearer understanding of algorithmic performance in complex and evolving environments.

The main contributions of this study can be summarized as follows:

- A unified experimental framework is developed to compare deterministic algorithms (Dijkstra and A*) and an evolutionary approach (GA) under identical routing conditions.
- Fixed layover constraints are explicitly incorporated into the route optimization problem, addressing a limitation rarely considered in existing VRP and shortest-path studies.
- A GA tailored to airline networks is designed using path-based encoding, tournament selection, crossover, mutation, and clearly defined termination criteria.
- The GA configuration was systematically defined (population size = 200, mutation rate = 0.1, tournament size = 5, maximum generations = 100, stagnation-based stopping), ensuring reproducibility and fair comparison.
- A multi-constraint optimization model is introduced by integrating air time, passenger costs, and airline operational costs into a composite fitness function.
- A dynamic evaluation of network resilience was performed by analyzing the impact of removing and introducing direct routes in a real-world flight network.

The remainder of this paper is organized as follows. Section II reviews related work and identifies key research gaps. Section III details the methodology, including the data preprocessing, constraint formulation, and algorithmic implementation. Section IV presents the experimental results, followed by a comparative performance analysis in

Section V. Section VI concludes the study with key findings and recommendations for future research.

II. LITERATURE REVIEW

A wide spectrum of approaches has been proposed to solve the vehicle routing problem (VRP), ranging from classical deterministic algorithms to metaheuristics and heuristic search strategies. Within this broad landscape, three algorithmic paradigms have recently recurred in the literature: evolutionary algorithms, represented primarily by genetic algorithms (GAs); classical shortest-path algorithms, exemplified by Dijkstra's algorithm; and heuristic search algorithms, such as A* Search. These methods embody fundamentally different optimization philosophies and collectively provide a representative basis for examining how algorithmic design influences performance, scalability, and adaptability in complex routing environments. Although numerous alternative approaches, including Tabu Search and Ant Colony Optimization, have achieved notable results, the prevalence, conceptual diversity, and complementary characteristics of GAs, Dijkstra, and A* Search justify their selection for closer examination.

A substantial body of research has highlighted the flexibility and effectiveness of GAs for addressing a wide range of VRP. Early work demonstrated that adaptive genetic operators and the dynamic adjustment of crossover and mutation processes can significantly reduce routing costs in stochastic VRP formulations, although scalability becomes increasingly problematic as the network size increases [2]. Other studies have explored energy-efficient routing in heterogeneous vehicle fleets, showing that advanced genetic encodings and specialized operators outperform classical optimization techniques in minimizing energy consumption [3]. GAs have also been successfully applied to public transportation, such as university shuttle routing, where enhanced crossover mechanisms yield measurable reductions in travel distance and operational cost [4]. More recent research has combined GA optimization with machine-learning-based travel time prediction to improve accuracy in dense urban networks, demonstrating the strong potential of hybrid approaches to reduce cost and improve predictive performance [5]. Domain-specific adaptations have further extended GAs to specialized VRP variants, including tourism routing under time-window constraints [6] and dynamic VRPs involving real-time changes in network parameters [7]. Enhanced GA operators have also been proposed for the Travelling salesman problem to improve solution quality through specialized crossover and mutation strategies, although with limited generalizability to more complex routing problems [8]. Environmental sustainability has motivated the development of GA-based routing models designed to minimize CO2 emissions while maintaining operational feasibility [9]. Hybrid GAs have been further explored for multi-depot routing problems, such as those incorporating temporal-spatial clustering for improved cost efficiency [10] and urban waste collection

systems, where emissions and the total distance were effectively reduced [11]. Although these studies collectively demonstrate the versatility of GAs, they also underscore key limitations: hybridization increases algorithmic complexity, parameter tuning remains problem dependent, and scalability issues persist in large, high-dimensional routing spaces.

Classical deterministic algorithms continue to serve as essential baselines in routing research, with Dijkstra's algorithm being the most widely referenced for shortest-path computations [12]. Although effective for static and single-objective settings, Dijkstra's algorithm struggles in multi-constraint or dynamic VRPs. Attempts to improve scalability, such as integrating Dijkstra's algorithm with K-means clustering to simplify the CVRP into smaller route groups, showed reductions in computational time but a limited ability to address multi-objective demands [13]. Dynamic extensions incorporating real-time traffic predictions have also been proposed, which enable more responsive routing in large urban transportation networks [14]. However, even these adaptive variants remain constrained in large-scale airline networks where feasible paths multiply rapidly under realistic operational constraints.

Heuristic search algorithms, particularly A* Search, have also played an influential role in VRP research. A* Search enhances search efficiency by integrating heuristics that guide exploration, making it suitable for real-time or dynamic routing scenarios. Applications combining A* Search with GA optimization for electric vehicle routing with time windows demonstrated reduced energy consumption and improved delivery adherence [15]. By leveraging A*'s ability to search efficiently and combine it with GAs for optimization, they achieved superior results in reducing energy consumption and meeting delivery time requirements [15]. Additional work in autonomous vehicle navigation expanded A* Search to incorporate turning penalties and obstacle-avoidance heuristics, thereby improving the accuracy in dense urban networks [16]. These studies highlight the advantages of heuristic search but also emphasize its dependency on domain-specific heuristics and limited capacity to generalize across heterogeneous VRP settings without hybridization [16].

In addition to the methods discussed above, other meta-heuristic approaches have been widely applied to routing and optimization problems. Ant Colony Optimization (ACO), inspired by the foraging behavior of ants, has demonstrated strong performance in path finding tasks through pheromone-based learning and collective exploration of solution spaces [17]. Similarly, Particle Swarm Optimization (PSO) has been successfully applied to both continuous and combinatorial optimization problems by leveraging population-based search mechanisms and adaptive information sharing. Although these approaches have shown promising results in various VRP-related applications, they were not included in the present study to maintain a focused comparison between fundamentally different optimization

paradigms: deterministic shortest-path algorithms and evolutionary methods [18].

Several consistent trends have emerged across these diverse strands of work. Hybridization dominates recent methodological developments, with many studies combining evolutionary, deterministic, and machine learning components to overcome the limitations of individual algorithms. Scalability persists as a significant challenge, as algorithms that perform well on small or synthetic datasets often degrade sharply in real-world networks characterized by high dimensionality and dynamic constraints. Similarly, although VRP research commonly addresses capacity restrictions, time windows, energy constraints, or stochastic demand, explicit modeling of layover or stopover requirements, central to airline routing applications, remains largely absent. Few studies have evaluated how optimized routing solutions respond to structural network changes, such as the removal or introduction of routes, thus limiting the understanding of network resilience. Moreover, comparative analyses across fundamentally distinct algorithmic paradigms are rare, and existing studies typically focus on incremental improvements within a single methodological class.

These gaps motivated this study. By evaluating Dijkstra, A* Search, and a GA under a unified experimental framework, this study provides a cross-paradigm comparison that has rarely been explored in the existing VRP literature. The incorporation of explicit layover constraints addresses a neglected yet operationally important aspect of airline routing. Furthermore, by examining how algorithms respond to the removal and addition of direct routes, this study contributes new insights into network resilience and adaptability, two critical factors for understanding the optimization performance in real-world transportation systems. Through this integration of methodological diversity, realistic constraints, and perturbation analysis, the present study advances the current knowledge on route optimization in large-scale and dynamic networks.

III. METHODOLOGY

This section outlines the methodological steps taken to perform flight route optimization using three different algorithmic approaches: traditional shortest-path algorithms (Dijkstra's and A* search) and an evolutionary algorithm (Genetic Algorithm). The methodology is divided into two main phases: data preparation and exploratory data analysis (EDA), and the implementation and customization of optimization algorithms to fit the specific constraints of the problem.

A. DATA PREPARATION AND EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted as a foundational step to transform the raw flight dataset obtained from Kaggle [17] into a structured format suitable for optimization tasks. This phase aims to extract relevant features, remove

redundancy, handle unreliable data, and generate insights that support the subsequent comparison of the optimization algorithms. A visual overview of the entire EDA workflow is shown in Fig. 1, which provides a step-by-step summary of the raw dataset for the final insights.

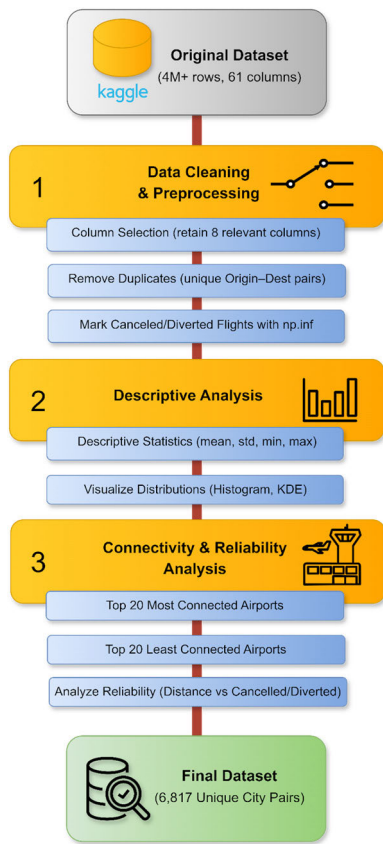


FIGURE 1. Data preprocessing and analysis workflow for flight route optimization.

To ensure dataset consistency, only unique origin-destination pairs were retained. Duplicate entries reflecting repeated flights across dates were removed, significantly reducing redundancy while preserving the underlying transportation network. Canceled and diverted flights were retained, but $\text{AirTime} = \infty$ was assigned to prevent selection during the optimization. This ensured that unreliable flights remained part of the structural network while being algorithmically unattractive as candidate routes. After preprocessing, the final dataset contained 6,817 unique and reliable routes. Detailed exploratory analysis was conducted on the cleaned dataset.

Summary statistics indicated that most flights lasted between 1 and 2.5 hours, with an average air time of 123 min and a median of 107 min. The flight distance ranged from 31 to 5,095 miles, with a median of 776 miles. Both variables exhibited right-skewed distributions, which is consistent with typical patterns in real-world airline networks. The visualizations in Fig. 2 and 3 confirm that the majority of the routes are short- to medium-haul. Longer flights appear

less frequently and show higher rates of cancellation or diversion, which is a pattern relevant for optimization because it informs the design of fitness functions or penalty mechanisms for unreliable paths.

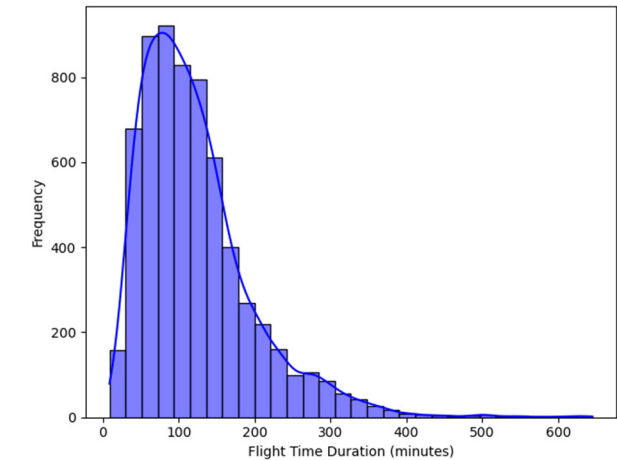


FIGURE 2. Flight air time distribution.

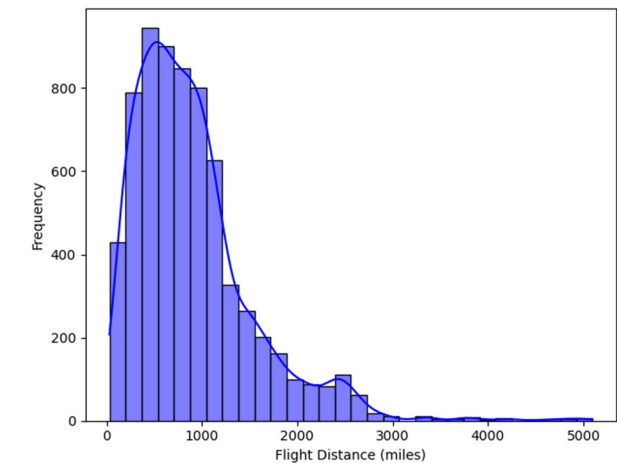


FIGURE 3. Distribution of flight distance.

The analysis also examined the airport connectivity. The 20 busiest airports and the least busy airports were identified based on their total number of direct connections, as summarized in Table 1.

The busiest airports, such as Dallas/Fort Worth (DFW) and Chicago O’Hare (ORD), have more than 200 connections, making them dominant network hubs. By contrast, many regional airports have only two direct connections, indicating significant sparsity in certain parts of the network. This sparsity has direct implications for the optimization process. Although large hubs offer numerous routing alternatives, poorly connected airports introduce structural bottlenecks that algorithms must address explicitly. An analysis of flights disrupted by distance showed that canceled and diverted flights were more frequent on shorter routes. This finding suggests that disruptions do not always correlate with

TABLE 1. Busiest airports and least busy airports.

Rank	Busiest Airports		Least Busy Airports	
	Airport Code	Connections	Airport Code	Connections
1	DFW	376	ABI	2
2	ORD	357	FSM	2
3	DEN	366	FLO	2
4	ATL	308	SCC	2
5	CLT	288	DHN	2
6	LAS	248	COD	2
7	IAH	240	MKG	2
8	MSP	236	CYS	2
9	PHX	216	DBQ	2
10	LAX	210	SJT	2
11	DTW	202	PPG	2
12	DCA	198	DIK	2
13	PHL	195	EKO	2
14	IAD	188	SHR	2
15	MCO	186	DLG	2
16	SEA	186	PQI	2
17	EWR	182	DRT	2
18	FLL	180	EAT	2
19	SLC	176	EAU	2
20	BNA	170	PGV	2

route length and should be considered in future optimization constraints. Overall, EDA ensured that the dataset used for optimization was clean, compact, and structurally reliable. Preprocessing improves the input quality while preserving network complexity. The insights gained from the connectivity patterns, flight reliability, and distance distributions provide a strong foundation for subsequent optimization models.

B. ALGORITHM IMPLEMENTATION

Following data preprocessing, the next phase involved implementing three algorithmic approaches: Dijkstra's algorithm, A* search algorithm, and Genetic Algorithm (GA). Each algorithm was adapted to operate on the processed dataset and was applied to solve the flight route optimization problem. These methods were selected because they represent two distinct optimization paradigms: deterministic and evolutionary. The objective was not only to identify efficient routes but also to evaluate algorithmic performance under realistic operational constraints, particularly the requirement of enforcing with an exact number of layovers. In total, six algorithmic configurations were implemented and organized into three categories.

- 1) Standard pathfinding algorithms:
 - a) Dijkstra's Algorithm
 - b) A* Search Algorithm with precomputed air-time heuristics
- 2) Customized pathfinding algorithms with exact layover constraints:
 - a) Custom Dijkstra's Algorithm with Exact Layovers
 - b) Customized A* Algorithm with Exact Layovers
- 3) Genetic Algorithms (GAs):
 - a) GA with air time as the objective
 - b) GA with multiple cost-related constraints

Each algorithm is briefly described below, with emphasis on how it was adapted to the dataset and problem-specific requirements.

1) STANDARD PATHFINDING ALGORITHMS

The standard Dijkstra's algorithm was used as the baseline for the shortest-path computation. Airports were modeled as nodes and flights as edges, with airtime serving as the edge weight. The A* Search algorithm integrates the following two cost components:

- g-score: actual travel cost from the origin to the current node
- h-score: estimated cost from the current node to the destination where the heuristic is defined as the precomputed shortest air-time between nodes

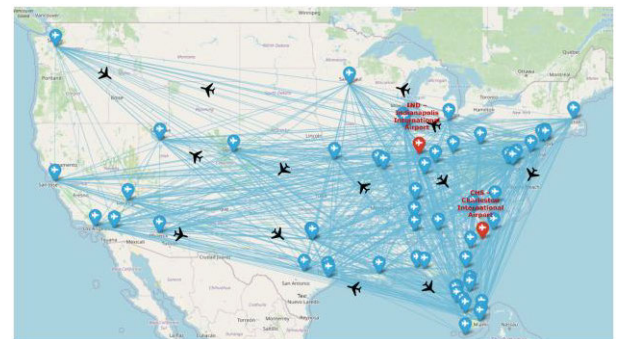
To improve efficiency, this study uses precomputed shortest-path air-time values obtained using the Floyd–Warshall algorithm, which calculates the minimum travel time between all airport pairs. These precomputed values serve as heuristics, significantly reducing the runtime during the search.

2) CUSTOMIZED PATHFINDING ALGORITHMS WITH EXACT LAYOVER CONSTRAINT

The customized Dijkstra's algorithm was adapted to enforce the exact number of required layovers. In this version:

- Each explored path tracked both cumulative cost and number of stops.
- The priority queue expanded only partial paths that remained consistent with the layover constraint.
- A path is accepted as a solution only if it satisfies the exact layover requirement.

The customized A* search incorporates the exact layover condition into its heuristic using a penalty mechanism. The heuristic cost increases when a candidate path deviates from the required number of layovers, thereby guiding the algorithm toward constraint-compliant solutions.

**FIGURE 4.** Visual representation of the flight network based on adjacency list structure.

3) GENETIC ALGORITHMS

For the GA to minimize the total air time, each individual in the population represents a completely feasible route between

the origin and destination airports. The initial population was generated using an adjacency-list structure derived from the dataset, ensuring that all initial routes were valid. A map generated from this adjacency list is shown in Fig. 4. Fitness was calculated as the total air time, with shorter routes receiving higher fitness. Tournament selection was used for parent selection, and offspring were produced through crossover and mutation. Elitism preserves the best routes across generations, guiding convergence toward efficient solutions.

For clarity, GA operates on path-based encoding, where each individual represents a complete route as an ordered sequence of airports. Parent solutions are selected using tournament selection, and new offspring are generated through a single-point crossover mechanism, that combines segments of two parent routes while preserving the feasibility. Mutation is applied by replacing one intermediate airport with a valid alternative based on the adjacency list, thereby ensuring that all the generated routes remain feasible. The algorithm runs for a fixed number of generations and may terminate early if no improvement is observed over a predefined number of iterations (stagnation criterion). The full set of parameter values used in all the experiments is provided in Table 3.

Before constructing the adjacency list, the dataset was validated to ensure consistency. Specifically, the algorithm verifies that both origin and destination airports exist in the dataset. If the origin airport is missing, an error arises because the search cannot be initiated. Similarly, if no valid connections lead to a destination airport, then the process is terminated to prevent invalid or infeasible route generation.

This version enabled a direct comparison with Dijkstra's and A* searches, both of which optimize the airtime. The second GA configuration extended the objective function to include both passenger- and airline-related costs, thereby producing a more realistic optimization model. The overall structure of the GA remains unchanged however the fitness function is expanded to incorporate the following components:

1. Passenger Cost

$$\begin{aligned} \text{passenger_cost} \\ &= (\text{distance} \cdot \text{passenger_cost_per_mile}) \\ &+ (\text{airtime} \cdot \text{passenger_cost_per_minute}) \end{aligned} \quad (1)$$

2. Airline Cost:

$$\begin{aligned} \text{airline_cost} \\ &= \text{fuel_price} \cdot (\text{fuel_rate_per_mile} \cdot \text{distance}) \\ &\cdot \text{load_factor} \end{aligned} \quad (2)$$

The equation estimates the operating cost per flight segment by adjusting for the load factor and proportion of occupied seats. A higher load factor reduces per-passenger costs and encourages efficient seat utilization.

3. Total Cost

$$\text{total_cost} = \text{passenger_cost} + \text{airline_cost} \quad (3)$$

The total cost served as the fitness score, where lower values indicate better solutions. The parameter values used in these equations were derived from credible aviation industry sources 2022 and are summarized in Table 2.

TABLE 2. Parameter values and references used in cost calculations.

Parameter	Value	Source / Justification
fuel_price	\$3.28	[18]
fuel_rate_per_mile	0.05 gal/seat/mile	[19]
load_factor	0.83	[20]
passenger_cost_per_mile	\$0.14	[21]
passenger_cost_per_minute	\$0.20	[22]

IV. RESULTS

This section presents an empirical evaluation of three algorithmic approaches for flight route optimization: Dijkstra's algorithm, A* search algorithm, and Genetic Algorithm (GA). The results address the performance of these methods under fixed layover constraints and analyze the behavior of the multi-constraint GA when the network structure is modified by removing or introducing direct routes. Owing to the large size of the outputs, detailed numerical results are provided in Appendices A–E.

A. COMPARISON OF DIJKSTRA, A*, AND GENETIC ALGORITHM WITH AIR TIME CONSTRAINT

To assess the baseline optimization performance, the algorithms were evaluated on identical origin–destination pairs with an imposed requirement for the exact number of layovers. These test cases include high-connectivity hubs (e.g., DFW, SEA, and LAX) and sparsely connected regional airports (e.g., ABI and DIK), enabling consistent comparisons across different network densities. The GA configurations used in all experiments are listed in Table 3. It should be noted that GA is inherently stochastic. In this study, the reported results corresponded to a single representative run for each experimental scenario. Although this approach allows for a consistent comparison with deterministic algorithms, it does not capture the variability across multiple runs.

These values were chosen to maintain a balance between performance and speed. Although the population size was set to 200, this parameter was effectively adapted to the dataset because its influence depended on the number of feasible routes in the network. In cases where the dataset provides fewer possible connections, the population size naturally behaves as an adaptive factor, and its diversity is constrained by the actual number of feasible route combinations available. Conversely, when more feasible paths exist, the same population size supports broader exploration without unnecessary computational overheads. Elitism maintains the best solutions, whereas mutation helps to discover new ones. Tournament selection provides fair opportunities for routes from less-connected airports. This configuration worked well with our dataset, which included 6,817 routes and exhibited

pronounced hub-and-spoke characteristics, as reflected in the connectivity imbalance summarized in Table 1.

The selection of GA parameters is closely related to the heterogeneous connectivity structure of the airline network. As shown in Table 1, the dataset exhibits a strong imbalance between highly connected hub airports (e.g., DFW, ORD, and DEN) and sparsely connected regional airports with only two direct connections. This structural asymmetry motivated the use of tournament selection and a moderate mutation rate, which helped preserve feasible routes originating from low-degree nodes while still allowing effective exploration around major hubs. The chosen population size ensures sufficient diversity in densely connected regions, without artificially inflating the search space for sparsely connected airports where the number of feasible route combinations is inherently limited. Consequently, the GA configuration is aligned with the empirical network topology rather than being arbitrarily tuned. All GA parameters were fixed across all experiments to ensure consistency and a fair comparison with deterministic algorithms under different network configurations and layover constraints.

TABLE 3. Configuration parameters for route optimization.

Parameter	Value
Population Size	200
Elitism Percentage	0.1
Tournament Size	5
Max Generations	100
Max Stagnation	50
Mutation Rate	0.1

The results obtained by running Dijkstra's algorithm are listed in Table 4 (see Appendix A). This table presents the computed routes for each test case, showing the shortest air-time route when considering a fixed number of layovers. The results obtained from A* are presented in Table 5 (see Appendix B). Similar to Dijkstra's algorithm, this table presents the computed routes considering the different layover constraints. The results obtained from the Genetic Algorithm with the AirTime constraint are summarized in Table 6 (see Appendix C). This table follows the same structure as the previous ones, displaying the computed routes for the different test cases.

B. EVALUATING GENETIC ALGORITHM WITH MULTIPLE COST CONSTRAINTS

To examine the algorithmic robustness in dynamic environments, the multi-constraint GA was tested under two structural perturbations: (1) removal of an existing direct route and (2) introduction of a new direct route. In all cases, the GA applied a composite cost function combining passenger cost, airline cost, and air time, as described in Section III.

1) IMPACT OF ROUTE REMOVAL ON NETWORK EFFICIENCY

For airport pairs with existing direct flights, the baseline route cost is computed using GA's multi-constraint fitness

function. The corresponding direct edge is then removed from the network, after which the GA identifies the most cost-efficient alternative path. The numerical results are presented in Table 7 (Appendix D).

2) IMPACT OF INTRODUCING A NEW DIRECT ROUTE

The second test examined airport pairs that had no direct flights. The GA was first used to determine the best indirect route based on cost and time. Next, a hypothetical direct flight is introduced between the two airports, and the cost of this new route is calculated. This test evaluates whether adding a direct flight results in a meaningful cost reduction or whether the current indirect routes already provide efficient alternatives. If the difference in costs is large, a new route would be useful. Otherwise, the current indirect route would be already efficient. Detailed results are shown in Table 8 (Appendix E).

V. DISCUSSION

This section interprets the empirical findings in relation to the research questions and situates them within the broader context of work on airline network optimization. The analysis focuses on how algorithmic behavior changes under fixed layover constraints and structural modifications of the flight network. These conditions reflect realistic operational scenarios, in which airlines must adhere to scheduling requirements while managing evolving route configurations. The results revealed clear patterns in the algorithmic efficiency, adaptability, and structural properties that determine network resilience.

Experiments that imposed an exact number of layovers demonstrated fundamental differences between the traditional shortest-path algorithms and evolutionary approaches. Dijkstra's algorithm consistently produces a true minimum-air-time route when a feasible solution exists. However, its performance deteriorated rapidly as the number of required layovers increased, particularly in the sparsely connected regions of the network. The enforced constraint enlarges the search space in combination, leading to a substantial growth in runtime. In cases where no valid path existed, Dijkstra traversed nearly the entire graph before concluding the infeasibility. These outcomes align with the literature, which emphasizes Dijkstra's reliability but also highlights its limited scalability under complex or combinatorial constraints. A* search improves the computation time in highly connected subgraphs by leveraging its heuristic to reduce node exploration. However, the strict layover constraint exposes a structural limitation: although the heuristic is aligned with the true optimization objective, it does not explicitly account for the exact layover requirement. Consequently, an A* search may occasionally converge to suboptimal routes or fail to identify feasible solutions under strict constraints. Prior research indicates that A* performs effectively when the heuristic closely approximates the true remaining cost; and the present results reinforce this observation, while also demonstrating that even well-aligned heuristics can become

TABLE 4. Results of Dijkstra's algorithm.

Test ID	Origin Airport	Destination Airport	Exact Layovers	Direct Flight	AirTime (min)	Route	Execution Time
1	DFW	SEA	0	Yes	239.0	['DFW', 'SEA']	0.0160
2	DFW	SEA	2	Yes	243.0	['DFW', 'DEN', 'SEA']	0.0772
3	DFW	SEA	3	Yes	255.0	['DFW', 'COS', 'DEN', 'SEA']	1.0967
4	DFW	SEA	4	Yes	271.0	['DFW', 'COS', 'DEN', 'BZN', 'SEA']	9.3011
5	DFW	SEA	5	Yes	298.0	['DFW', 'COS', 'DEN', 'CNY', 'SLC', 'SEA']	100.2430
6	MSP	ONT	0	No	199.0 (est.)	No direct path	N/A
7	MSP	ONT	2	No	209.0	['MSP', 'DEN', 'ONT']	0.0050
8	MSP	ONT	3	No	218.0	['MSP', 'DEN', 'LAS', 'ONT']	0.5572
9	LAX	LGA	0	Yes	286.0	['LAX', 'LGA']	0.0056
10	LAX	LGA	2	Yes	287.0	['LAX', 'PIT', 'LGA']	0.0075
11	LAX	LGA	3	Yes	308.0	['LAX', 'ABQ', 'MCI', 'LGA']	1.0530
12	LAX	LGA	4	Yes	317.0	['LAX', 'COS', 'DEN', 'SYR', 'LGA']	8.0548
13	LAX	LGA	5	Yes	336.0	['LAX', 'COS', 'DEN', 'MCI', 'STL', 'LGA']	65.3014
14	ABI	DIK	0	No	166.0 (est.)	No direct path	N/A
15	ABI	DIK	2	No	N/A	No valid path was found from ABI to DIK with exactly 2 layovers.	N/A
16	ABI	DIK	3	No	208.0	['ABI', 'DFW', 'COS', 'DEN', 'DIK']	0.2262
17	ABI	DIK	4	No	220.0	['ABI', 'DFW', 'COS', 'DEN', 'DIK']	0.2262
18	ABI	DIK	5	No	271.0	['ABI', 'DFW', 'AUS', 'AMA', 'DEN', 'DIK']	4.6524

less informative under additional structural constraints, such as fixed layovers. It is important to emphasize that these limitations are not inherent to the A* algorithm itself, but rather arise from the heuristic design in constrained settings. In this study, an air-time-based heuristic was employed, where the remaining cost was estimated using the precomputed shortest air-time values obtained via the Floyd–Warshall algorithm. Although this approach ensures strong alignment with the objective function, it introduces additional preprocessing overhead and reduces the practical distinction between A* and Dijkstra's algorithm, because both rely on globally optimal path information, although A* retains its guided search structure.

From a scalability perspective, these behaviors become more pronounced as the network grows in size and heterogeneity. In large airline graphs, the number of feasible paths increases rapidly with both the number of airports and imposed layover requirement, leading to a combinatorial expansion of the constraint-compliant search space. For Dijkstra's algorithm and A* search, this expansion translates into increased node and state exploration, particularly when exact layover constraints force the algorithms to enumerate longer partial paths before reaching feasibility (or proving infeasibility). In contrast, Genetic Algorithm exhibits a more stable computational footprint, because its runtime is primarily governed by the population size and number of generations rather

TABLE 5. Results of A* algorithm.

Test ID	Origin Airport	Destination Airport	Exact Layovers	Direct Flight	AirTime (min)	Route	Execution Time
1	DFW	SEA	0	Yes	239.0	['DFW', 'SEA']	0.0010
2	DFW	SEA	2	Yes	243.0	['DFW', 'DEN', 'SEA']	0.0009
3	DFW	SEA	3	Yes	776.0	['DFW', 'COS', 'DEN', 'SEA']	0.0240
4	DFW	SEA	4	Yes	1155.0	['DFW', 'RNO', 'OAK', 'EUG', 'SEA']	0.1091
5	DFW	SEA	5	Yes	N/A	No valid path found from DFW to SEA with exactly 5 layovers.	N/A
6	MSP	ONT	0	No	199.0 (est.)	No direct path	N/A
7	MSP	ONT	2	No	209.0	['MSP', 'DEN', 'ONT']	0.0050
8	MSP	ONT	3	No	625.0	['MSP', 'PHX', 'LAS', 'ONT']	0.6670
9	LAX	LGA	0	Yes	286.0	['LAX', 'LGA']	0.0011
10	LAX	LGA	2	Yes	443.0	['LAX', 'PIT', 'LGA']	0.0058
11	LAX	LGA	3	Yes	708.0	['LAX', 'PIT', 'IAD', 'LGA']	0.0099
12	LAX	LGA	4	Yes	1078.0	['LAX', 'PIT', 'IAD', 'RIC', 'LGA']	0.0166
13	LAX	LGA	5	Yes	1776.0	['LAX', 'PIT', 'IAD', 'RIC', 'MCO', 'LGA']	0.0519
14	ABI	DIK	0	No	166.0 (est.)	No direct path	N/A
15	ABI	DIK	2	No	N/A	No valid path found from ABI to DIK with exactly 2 layovers.	N/A
16	ABI	DIK	3	No	758.0	['ABI', 'DFW', 'DEN', 'DIK']	0.0014
17	ABI	DIK	4	No	1164.0	['ABI', 'DFW', 'COS', 'DEN', 'DIK']	0.0082
18	ABI	DIK	5	No	1869.0	['ABI', 'DFW', 'COS', 'PHX', 'DEN', 'DIK']	0.0258

than the total number of feasible paths in the graph. This distinction clarifies why evolutionary optimization remained computationally steady in our experiments as the layover constraints intensified, whereas deterministic methods showed substantial runtime growth in the same network.

The Genetic Algorithm demonstrated a markedly different behavior. Although it does not guarantee optimality, it consistently produces near-optimal solutions while maintaining stable computational performance with increasing complexity. Unlike Dijkstra or A*, runtime of GA did not increase significantly as the layover requirements increased, and with a fixed population size and number of generations, the computational cost remained almost constant.

In scenarios requiring multiple layovers, particularly those involving low-degree nodes, the GA outperformed both deterministic algorithms, often identifying feasible and competitive solutions within a fraction of the time. These results resonate with those of prior studies that reported the robustness of evolutionary methods in large or irregular search spaces. However, the current findings extend this understanding by demonstrating the effectiveness of GA under the explicit structural constraints imposed on the route.

The evaluation of network resilience, addressed through the experimental removal of direct flights, provided further insight into the interplay between topology and optimization behavior. In sparsely connected areas, removing a direct route

TABLE 6. Results of the GA algorithm (air time constraint).

Test ID	Origin Airport	Destination Airport	Exact Layovers	Direct Flight	AirTime (min)	Route	Execution Time
1	DFW	SEA	0	Yes	239.00	['DFW', 'SEA']	11.3616
2	DFW	SEA	2	Yes	243.00	['DFW', 'DEN', 'SEA']	0.0982
3	DFW	SEA	3	Yes	259.00	['DFW', 'DEN', 'BZN', 'SEA']	0.2203
4	DFW	SEA	4	Yes	271.0	['DFW', 'COS', 'DEN', 'BZN', 'SEA']	0.3093
5	DFW	SEA	5	Yes	322.00	['DFW', 'AUS', 'AMA', 'DEN', 'BZN', 'SEA']	0.8369
6	MSP	ONT	0	No	199.0	No direct path	N/A
7	MSP	ONT	2	No	209.00	['MSP', 'DEN', 'ONT']	0.0398
8	MSP	ONT	3	No	218.00	['MSP', 'DEN', 'LAS', 'ONT']	0.0398
9	LAX	LGA	0	Yes	286.00	['LAX', 'LGA']	20.2058
10	LAX	LGA	2	Yes	287.0	['LAX', 'PIT', 'LGA']	0.0321
11	LAX	LGA	3	Yes	309.0	['LAX', 'OMA', 'STL', 'LGA']	0.3480
12	LAX	LGA	4	Yes	331.0	['LAX', 'DEN', 'MCI', 'IAD', 'LGA']	0.0581
13	LAX	LGA	5	Yes	357.00	['LAX', 'PHX', 'ATL', 'CLT', 'BWI', 'LGA']	0.8289
14	ABI	DIK	0	No	166.0	No direct path	N/A
15	ABI	DIK	2	No	N/A	No valid path found from ABI to DIK with exactly 2 layovers	N/A
16	ABI	DIK	3	No	208.0	['ABI', 'DFW', 'DEN', 'DIK']	0.5093
17	ABI	DIK	4	No	220.0	['ABI', 'DFW', 'COS', 'DEN', 'DIK']	10.7637
18	ABI	DIK	5	No	441.00	['ABI', 'DFW', 'COS', 'ORD', 'DEN', 'DIK']	25.7532

produces substantial increases in travel time and cost, sometimes exceeding 200%, thereby illustrating the fragility of the low-redundancy regions. In contrast, major hubs exhibited negligible efficiency loss following route removal, owing to the abundance of alternative paths. Earlier VRP literature acknowledges the importance of redundancy, but generally examines static networks, and the present findings highlight that the magnitude of disruption is strongly conditioned by node degree and local connectivity patterns.

The introduction of hypothetical direct routes produces asymmetric effects across the different network regions. When the existing connections were highly indirect, the new route significantly reduced the total cost by over 40%,

underscoring the value of alleviating the structural bottlenecks. However, when indirect routes were already short and well-supported by nearby hubs, adding a direct link yielded marginal gains. These outcomes reflect the principles found in air transportation network studies, yet the integration of evolutionary optimization and layover-aware modeling provides a more operationally grounded perspective: a direct connection is most beneficial when the current network forces inefficient detours, not only when two airports lack a direct flight.

Taken together, these observations offer a broader synthesis of the algorithmic performance under realistic constraints. Deterministic algorithms are advantageous for small or

TABLE 7. Genetic algorithm performance after route removal.

Origin	Destination	Direct Cost	GA Route	GA Cost	Cost Difference (%)
JFK	EGE	605.92	[JFL, 'DEN', 'EGE']	577,4510	-4.698475
ATL	LEX	107.48	[ATL, CLT, LEX]	172,3045	60.313082
BNA	PSP	594.76	[BNA, PHX, PSP]	564,6350	-5.065068
IAH	PHX	350.88	[IAH, ELP, PHX]	334,6090	-4.637198
IAD	PDX	804.44	[IAD, MDW, PDX]	764,7680	-4.931629
LAX	CLT	729.40	[LAX, SGF, CLT]	687,9485	-5.682959
CID	DTW	150.40	[CID, ORD, DTW]	148,9985	-0.931848
DFW	BNA	219.12	[DFW, XNA, BNA]	232.4610	6.088445
MCI	SAN	462.68	[MCI, PHX, SAN]	442,6380	-4.331720
AMA	AUS	145.16	[AMA, DFW, AUS]	167,3370	-15.277625
HRL	MCO	351.92	[HRL, HOU, MCO]	365,2310	3.782394
MIA	LAX	813.04	[MIA, MSY, LAX]	770.2575	-5.262041
SBN	DTW	57.04	[SBN, GRR, DTW]	73.6785	29.169881
PIA	DFW	235.24	[PIA, ORD, DFW]	314.4985	33.692612
ATL	AEX	177.00	[ATL, JAN, IAH, AEX]	299.6170	69.275141
DEN	LGA	557.08	[DEN, DSM, LGA]	526.2200	-5.661527
MSO	DFW	451.02	[MSO, DEN, DFW]	430.9200	-4.494681
MSP	CWA	63.00	[MSP, ORD, CWA]	187.0405	196.896032
BDL	LAS	799.64	[BDL, MDW, LAS]	752.9630	-5.837252
LAX	BZN	314.44	[LAX, SLC, BZN]	31.2595	-1.011481
DEN	CVG	367.48	[DEN, MDW, CVG]	373.2640	1.573963
TUS	PHX	41.20	[TUS, LAS, PHX]	206.7200	401.747573
SDF	LAS	563.28	[SDF, DEN, LAS]	541.1120	-3.935521
LAX	PBI	800.00	[LAX, IAH, PBI]	755.8225	-5.522188
PWM	BNA	354.20	[PWM, EWR, BNA]	343.1420	-3.121965

TABLE 7. (Continued.) Genetic algorithm performance after route removal.

ANC	SLC	728.80	[ANC, SEA, SLC]	696.9595	-4.368894
DFW	ATL	252.12	[DFW, BHM, ATL]	242.5485	-3.796406
PWM	DEN	619.44	[PWM, MDW, DEN]	589.6150	-4.814833
SLC	BNA	479.88	[SLC, DEN, BNA]	455.1175	-5.160144
GSO	ATL	107.92	[GSO, CLT, ALT]	108.6915	0.714881

densely connected networks, where optimality and predictability are essential. A* is effective when heuristics align with cost structures; however, its reliability decreases under nonstandard constraints. The GA consistently performs well in dynamic, heterogeneous, and constrained environments, indicating its superior adaptability to airline network conditions characterized by uneven connectivity, fluctuating topology, and operational restrictions. These findings support the view echoed in recent hybrid routing studies that evolutionary methods offer a scalable alternative in contexts where classical algorithms encounter structural or computational limitations [2].

The study's interpretation also acknowledges the methodological boundaries. The cost model incorporates fixed estimates of fuel consumption, load factor, and passenger time. In practice, these values fluctuate with market conditions, aircraft types, weather, and operational constraints. Furthermore, real airline planning must account for slot availability, maintenance cycles, crew schedules, stochastic delays, and factors that are not modeled here. The optimization remains single-route focused rather than network-wide, and the GA, while adaptable, does not integrate local improvement heuristics that can accelerate the convergence. These limitations suggest several directions for future work, including dynamic cost modeling, incorporation of operational constraints, multi-objective formulations balancing passenger and airline priorities, and hybrid evolutionary–deterministic methods. In addition, owing to the stochastic nature of GA, the results reported in this study were based on single representative runs for each experimental scenario. Although this enables a consistent comparison with deterministic methods, it does not capture variability across multiple executions. Future work will incorporate repeated runs and statistical analysis to evaluate the solution stability and robustness.

Overall, this study contributes to the literature by offering a comparative, constraint-aware, and dynamically tested evaluation of route-optimization methods. It extends prior research by showing how algorithmic behaviors shift under structural changes, and how network topology interacts with optimization performance. The findings indicate that evolutionary algorithms, particularly GA, are well-suited for optimizing routes in large, irregular, and evolving airline networks,

where traditional shortest-path methods face inherent limitations. These insights are relevant both for advancing research in large-scale transportation optimization and for informing practical airline planning strategies aimed at improving network efficiency and resilience.

VI. CONCLUSION

This study investigated flight route optimization in a large real-world airline network by evaluating three algorithmic paradigms Dijkstra's algorithm, the A* search algorithm, and a Genetic Algorithm (GA) under realistic operational constraints and structural network modifications. The experimental design addressed three research questions concerning algorithmic efficiency under fixed layover requirements (RQ1), algorithmic behavior in response to route additions and removals (RQ2), and comparative adaptability across deterministic and evolutionary approaches (RQ3).

With respect to RQ1, the results demonstrate that deterministic shortest-path algorithms remain reliable for well-connected, low-complexity routing tasks but scale poorly when exact layover constraints significantly enlarge the feasible search space. Dijkstra's algorithm consistently returned optimal solutions, yet exhibited significant increases in computation time as the constraints intensified. A* achieved faster convergence in dense subgraphs, but proved sensitive to constraint-induced heuristic limitations under strict layover conditions, occasionally failing to identify valid solutions. In contrast, the GA maintained a stable computational performance and consistently generated near-optimal routes across heterogeneous connectivity patterns, highlighting its suitability for constrained and structurally irregular environments.

RQ2 examined the network resilience under structural perturbations through the removal and introduction of direct flight routes. The findings indicate that efficiency losses from route removal are strongly dependent on local connectivity; airports with low node degrees experience substantial increases in total costs, whereas dense hub regions absorb disruptions with minimal effects. Conversely, introducing hypothetical direct connections yielded pronounced efficiency gains when existing indirect routes were structurally inefficient but offered only marginal improvements when adequate alternatives already existed. These outcomes reinforce the

TABLE 8. Genetic algorithm performance after adding new routes.

Origin	Destination	GA Route	GA Cost	Direct Cost	Cost Difference (%)
PDX	RST	['PDX', 'MSP', 'RST']	489.34	478.4995	-2.265520
ACK	HHH	['ACK', 'PHL', 'HHH']	299.56	286.9750	-4.385399
ACV	DSM	['ACV', 'DEN', 'DSM']	528.25	563.6250	6.276336
MLI	MKE	['MLI', 'ORD', 'MKE']	73.71	62.7750	-17.419355
XNA	JAN	['XNA', 'DFW', 'JAN']	228.43	118.0500	-93.502753
AZA	AVL	['AZA', 'HOU', 'AVL']	596.08	512.1000	-16.399141
SDF	LAN	['SDF', 'DTW', 'LAN']	129.53	111.8000	-15.858676
HLN	CLE	['HLN', 'DEN', 'CLE']	583.83	514.6000	-13.453168
GTR	TPA	['GTR', 'ATL', 'TPA']	216.39	167.0750	-29.516684
MSN	IAD	['MSN', 'DTW', 'IAD']	232.94	237.9500	2.105484
STL	CIU	['STL', 'DTW', 'CIU']	241.99	245.4500	1.409656
TUL	CID	['TUL', 'ORD', 'CID']	257.47	144.9000	-77.688061
ACK	PVD	['ACK', 'LGA', 'PVD']	125.26	35.9800	-248.137854
SJC	AGS	['SJC', 'DFW', 'AGS']	751.32	727.5500	-3.267129
PWM	LAN	['PWM', 'EWR', 'DTW', 'LAN']	287.55	276.0500	-4.165912
IAD	FAY	['IAD', 'CLT', 'FAY']	150.64	110.5500	-36.264134
ECP	CLE	['ECP', 'ATL', 'CLE']	262.79	273.5500	3.933467
MDT	RIC	['MDT', 'IAD', 'RIC']	69.94	76.2000	8.215223
PNS	PDX	['PNS', 'DEN', 'PDX']	714.39	742.5500	3.792337
GRR	RAP	['GRR', 'MSP', 'RAP']	299.52	336.0000	10.857143
SAV	MOT	['SAV', 'MSP', 'MOT']	504.78	585.8000	13.830659
SBA	PSP	['SBA', 'LAX', 'PSP']	72.91	79.9500	8.805503
ONT	GEG	['ONT', 'LAS', 'GEG']	332.13	357.8500	7.187369
ICT	ANC	['ICT', 'DEN', 'ANC']	926.14	979.8500	5.481451
ABE	TTN	['ABE', 'IAD', 'RDU', 'TTN']	255.89	25.9250	-887.039537
HIB	ATL	['HIB', 'MSP', 'ATL']	356.27	362.8500	1.813422
ORF	BMI	['ORF', 'ORD', 'BMI']	280.24	264.8000	-5.830816
DEN	GCK	['DEN', 'DFW', 'GCK']	344.71	113.0500	-204.918178

TABLE 8. (Continued.) Genetic algorithm performance after adding new routes.

ICT	EWR	['ICT', 'STL', 'EWR']	413.98	430.3000	3.792703
KOA	CAK	['KOA', 'ORD', 'CAK']	1462.94	1410.7500	-3.699451
MSP	HOB	['MSP', 'DEN', 'HOB']	389.12	338.5000	-14.954210

importance of redundancy and structural balance in airline network designs.

RQ3 concerns the comparative adaptability of the evaluated algorithms. The GA demonstrated superior robustness across dynamic and highly constrained scenarios, effectively adapting to altered network topologies and producing high-quality solutions, where deterministic methods either became computationally infeasible or failed to identify feasible paths. These results suggest that evolutionary optimization is a strong candidate for real-world routing environments characterized by fluctuating connectivity, operational constraints, and large search spaces.

In addition to addressing these three research questions, this study contributes to the literature in several ways. It provides a unified, constraint-aware comparison of deterministic and evolutionary routing algorithms on a real-world airline network, which incorporates explicit layover constraints seldom examined in prior VRP research and offers a dynamic analysis of network resilience through the controlled removal and introduction of routes. Together, these contributions enhance our understanding of the interaction between topology, redundancy, and algorithm design in large-scale transportation systems.

This study has some limitations that suggest directions for future research. The cost model adopted here uses fixed parameters and does not incorporate dynamic market or operational factors such as fluctuating fuel prices, weather-induced delays, slot availability, or maintenance scheduling. Future research could integrate real-time data, develop multi-objective optimization frameworks that balance passenger and airline perspectives, or explore hybrid evolutionary-deterministic architectures to further improve solution quality and convergence speed.

In summary, the findings underscore the advantages of evolutionary optimization for complex, constraint-rich, and dynamically evolving airline networks while reaffirming the continued relevance of classical algorithms in simpler, well-structured scenarios. By demonstrating how algorithmic performance and network topology interact under realistic conditions, this study provides theoretical insights and practical guidance for designing resilient and efficient airline routing strategies.

APPENDIX A DETAILED RESULTS OF DIJKSTRA'S ALGORITHM

See Table 4.

APPENDIX B DETAILED RESULT OF A* ALGORITHM

See Table 5.

APPENDIX C DETAILED RESULTS OF GENETIC ALGORITHM (AIRTIME CONSTRAINT)

See Table 6.

APPENDIX D IMPACT OF ROUTE REMOVAL ON NETWORK PERFORMANCE

See Table 7.

APPENDIX E IMPACT OF ADDING NEW ROUTES TO THE NETWORK

See Table 8.

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